

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

Scenario: Web Development Standards Implementation

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B. <video src="video.mp4" controls></video>

Justification:

- The <video> element is the standard HTML5 tag used to embed videos in a webpage.
- The controls attribute provides built-in play, pause, volume, and fullscreen controls.
- It is supported by all modern web browsers.

Why the other options are incorrect:

- **A. <media>** – Not a valid HTML element.
- **C. <movie>** – Not a valid HTML5 tag.
- **D. <vid>** – Not a valid HTML tag.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

Justification:

- POST sends form data in the **request body**, making it more secure than GET.
- It is suitable for registration forms because it handles sensitive information like usernames, passwords, and email addresses.
- It supports large amounts of data and does not expose the data in the URL.

Why the other options are incorrect:

- **A. GET** – Sends data in the URL; not suitable for sensitive information.
- **C. PUT** – Used to update existing resources.
- **D. DELETE** – Used to delete resources.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. <nav>

Justification:

- The <nav> element represents a section containing navigation links.
- It improves website structure, accessibility, and search engine optimization (SEO).
- Screen readers can easily identify navigation menus.

Why the other options are incorrect:

- **A. <div>** – Generic container with no semantic meaning.
- **C. <section>** – Groups related content but is not specifically for navigation.

- **D. <header>** – Represents introductory content and may contain navigation, but it is not the dedicated navigation element.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: B.

```
<note>
```

```
  <to>User</to>
```

```
  <from>Admin</from>
```

```
</note>
```

Justification:

- Every opening tag has a matching closing tag.
- The XML document is properly nested and well-formed.

Errors in the other options:

- **A.**
- `<note><to>User</to><from>Admin</from>`
 - Missing the closing `</note>` tag.
- **C.**
- `<note><to>User</from></note>`
 - Tags are mismatched (`<to>` is closed with `</from>`).
- **D.**
- `<note to="User"></note`
 - Missing the closing angle bracket (`>`) in the end tag, making it invalid XML.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP

Justification:

- **HTTP (HyperText Transfer Protocol)** is the standard protocol used for communication between web browsers and web servers.
- It enables the transfer of web pages, images, videos, and other web resources.
- For secure communication, **HTTPS (HTTP Secure)** is commonly used, which encrypts data using SSL/TLS.

Why the other options are incorrect:

- **A. FTP** – Used for file transfer between computers.

- **C. SMTP** – Used for sending emails.
- **D. SNMP** – Used for network monitoring and management.

Summary of Correct Answers

Question	Correct Option	Answer
1	B	<code><video src="video.mp4" controls></video></code>
2	B	POST
3	B	<code><nav></code>
4	B	<code><note><to>User</to><from>Admin</from></note></code>
5	B	HTTP (HTTPS for secure web communication)

Code of Conduct:

I A.ADARSH(192365002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: